

Act V: Different Stages

Market Structures

Module XIV: The Corporate District

Monopoly and Market Power



Previously in Marginal Falls

At the Farmers Market:

- ◆ Many sellers.
- ◆ Identical goods.
- ◆ Easy entry.
- ◆ No single seller controlled price.
- ◆ Each stall was a price taker.

Next: Lantern Island is different.

One Gate, One Seller

To enter Founders' Night, citizens need a bridge pass.

River-Gate is the only seller of bridge passes.

There is *no nearby ferry, no second bridge* and *walking around the river is impossible* tonight.



River-Gate faces the whole town's demand.

Monopoly

*A **monopoly*** is a market with:

One seller.

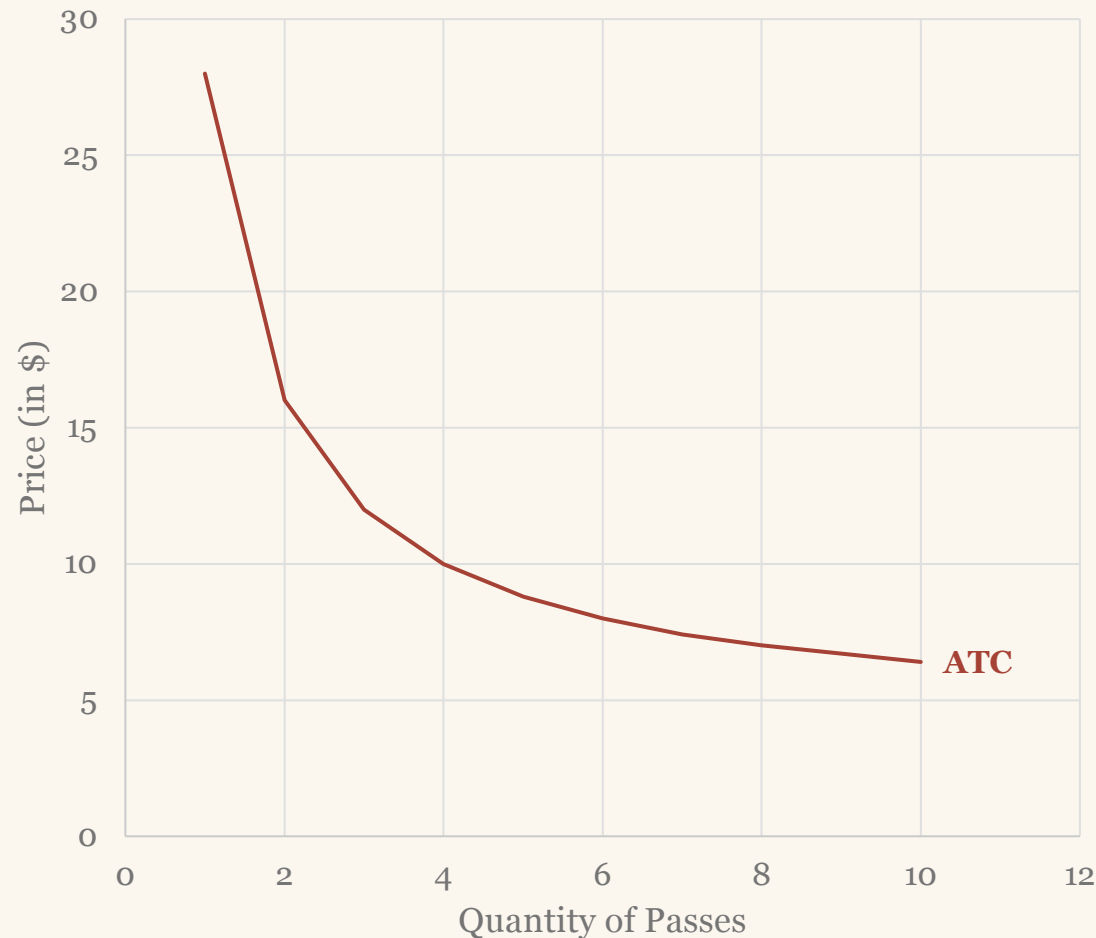
No close substitute.

Entry blocked.

River-Gate is not just “big”. River-Gate has market power.

Market power = ability to affect price.

Constructing the Bridge

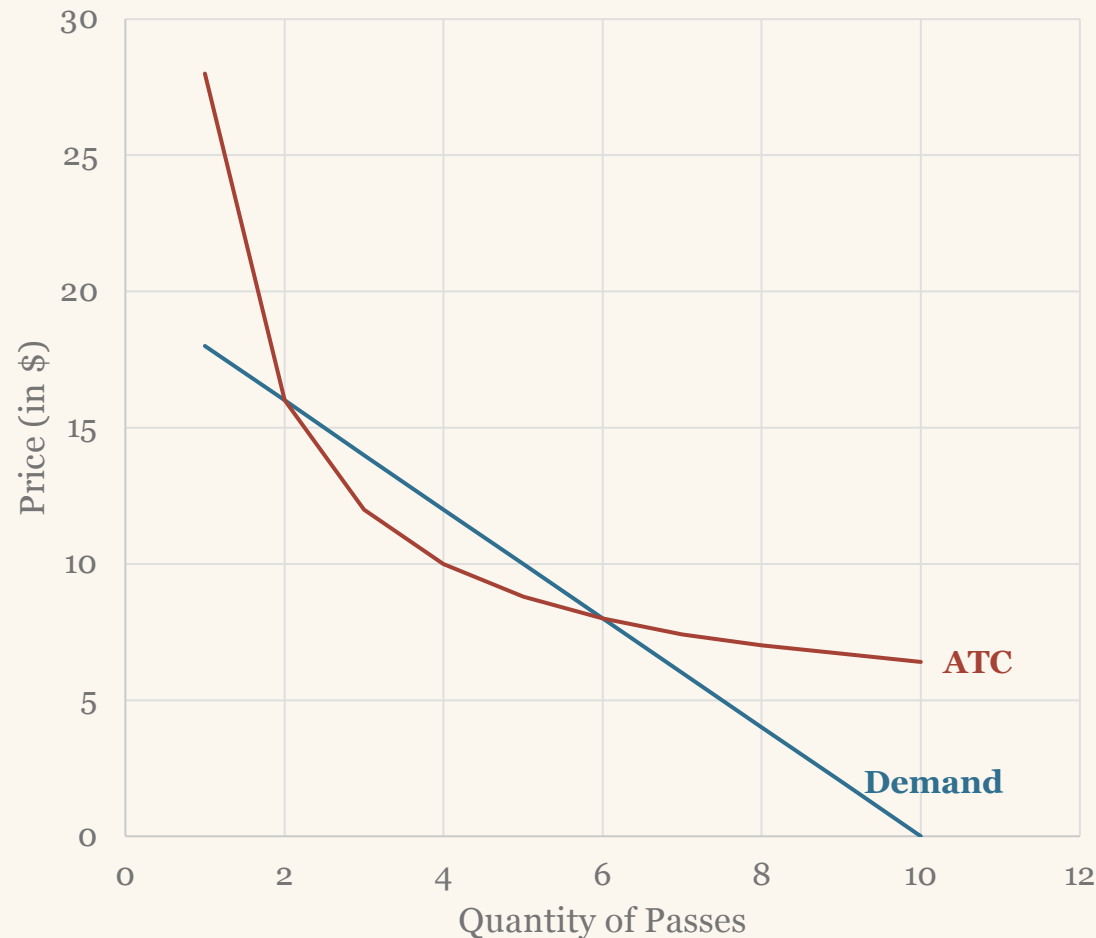


The bridge is expensive before anyone crosses.

Once the bridge exists, letting one more citizen cross is cheap.

So average cost can fall as more people use the bridge.

Constructing the Bridge



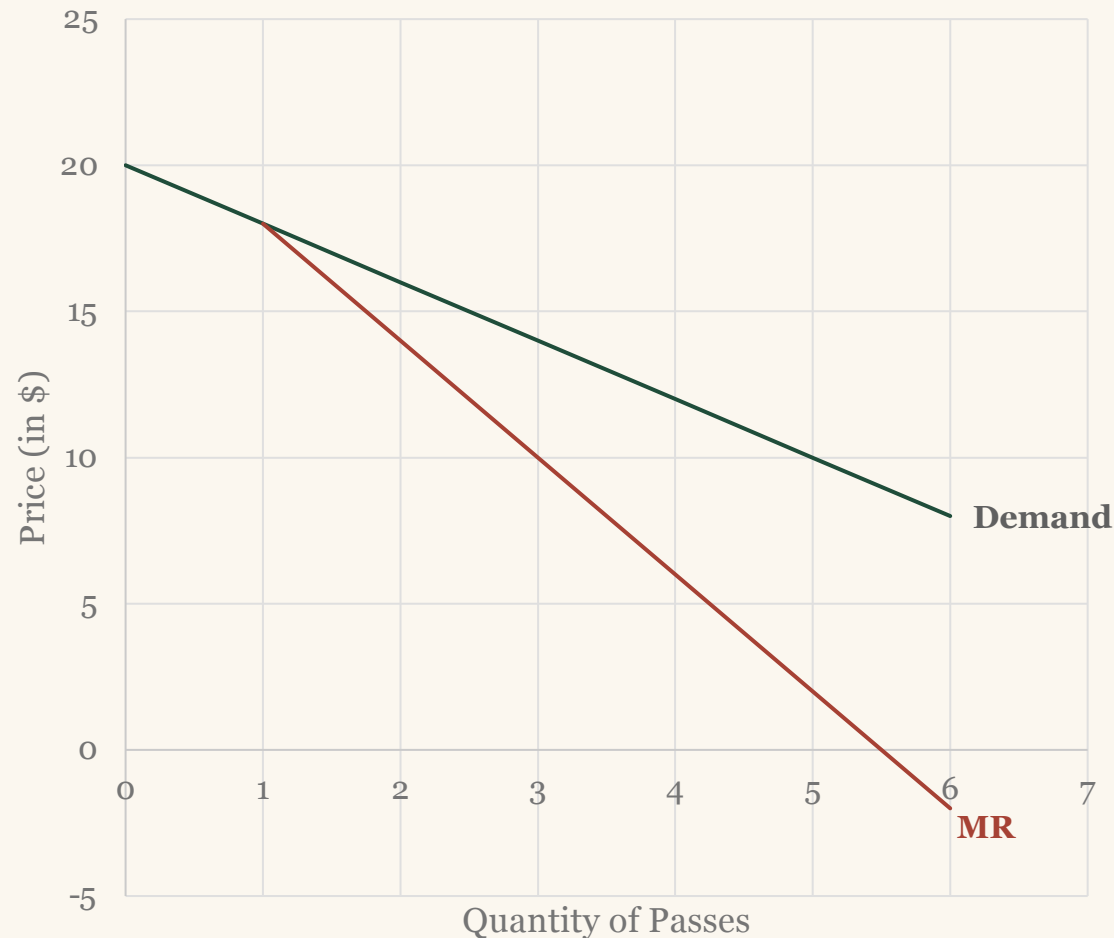
River-Gate asks:

How many passes can we sell at each price?

At a high price, only a few citizens enter.

At a lower price, more citizens enter.

One More Bridge Pass

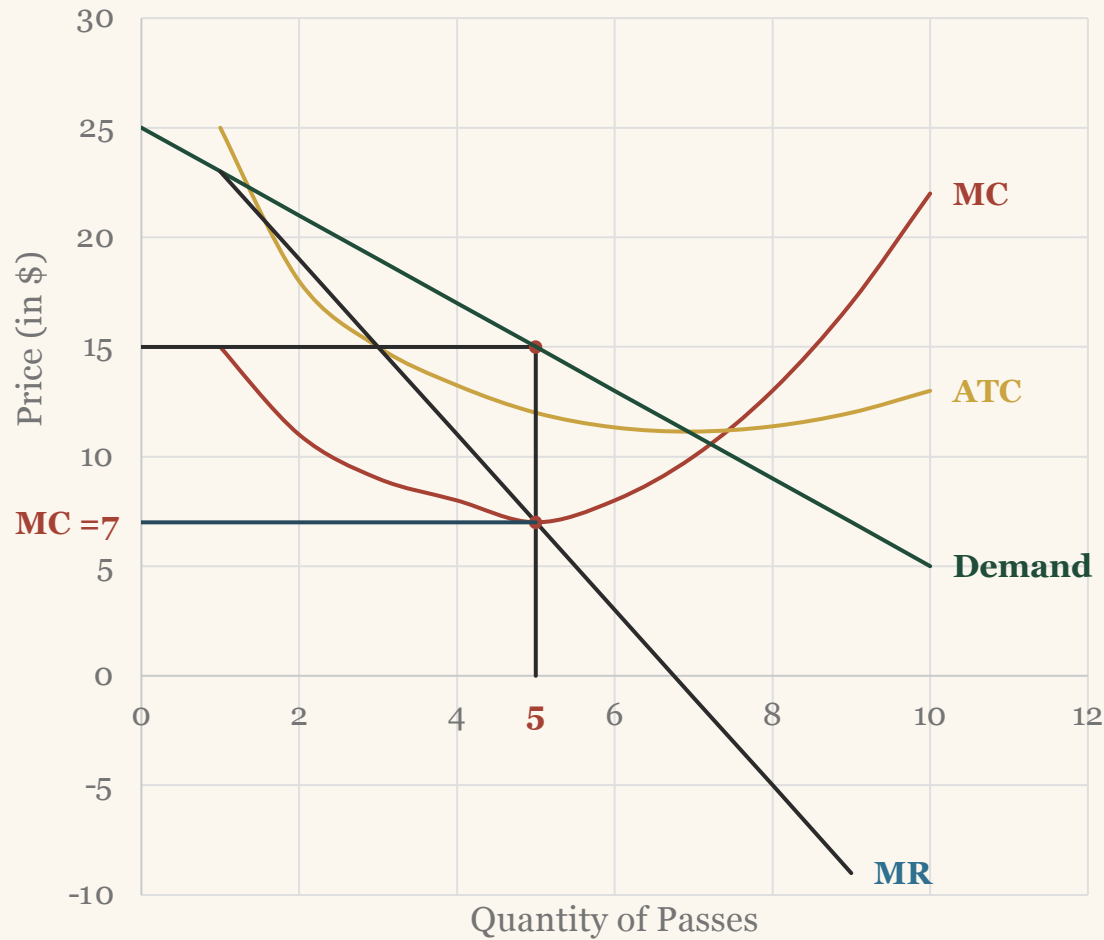


To sell more bridge passes:
River-Gate must lower the price.

That creates two effects:
Extra pass sold → revenue rises.
Old passes sell for less → revenue falls.

So, for a monopolist:
 $MR < Price$

Monopoly Output



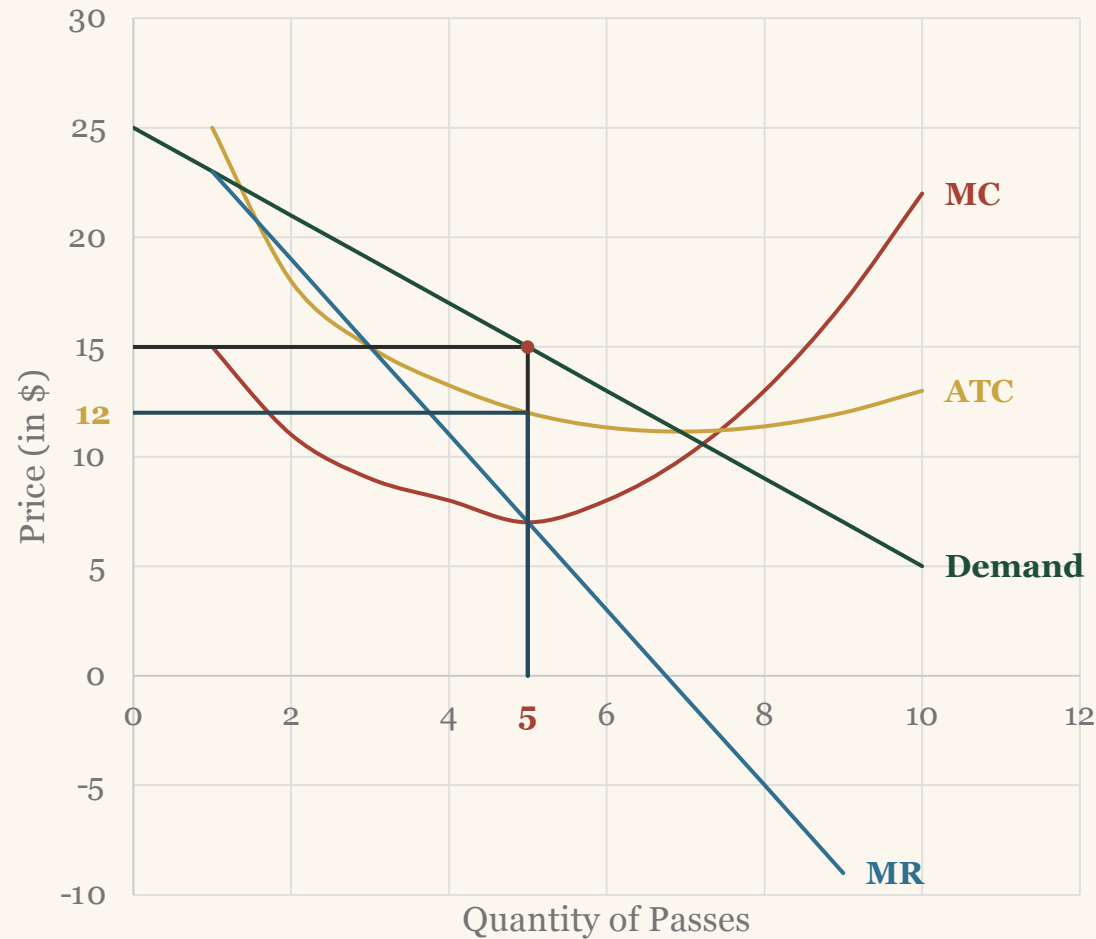
River-Gate chooses output where:

$$MR = MC$$

Optimal Quantity: 5 passes

Optimal Price: \$15

Monopoly Output

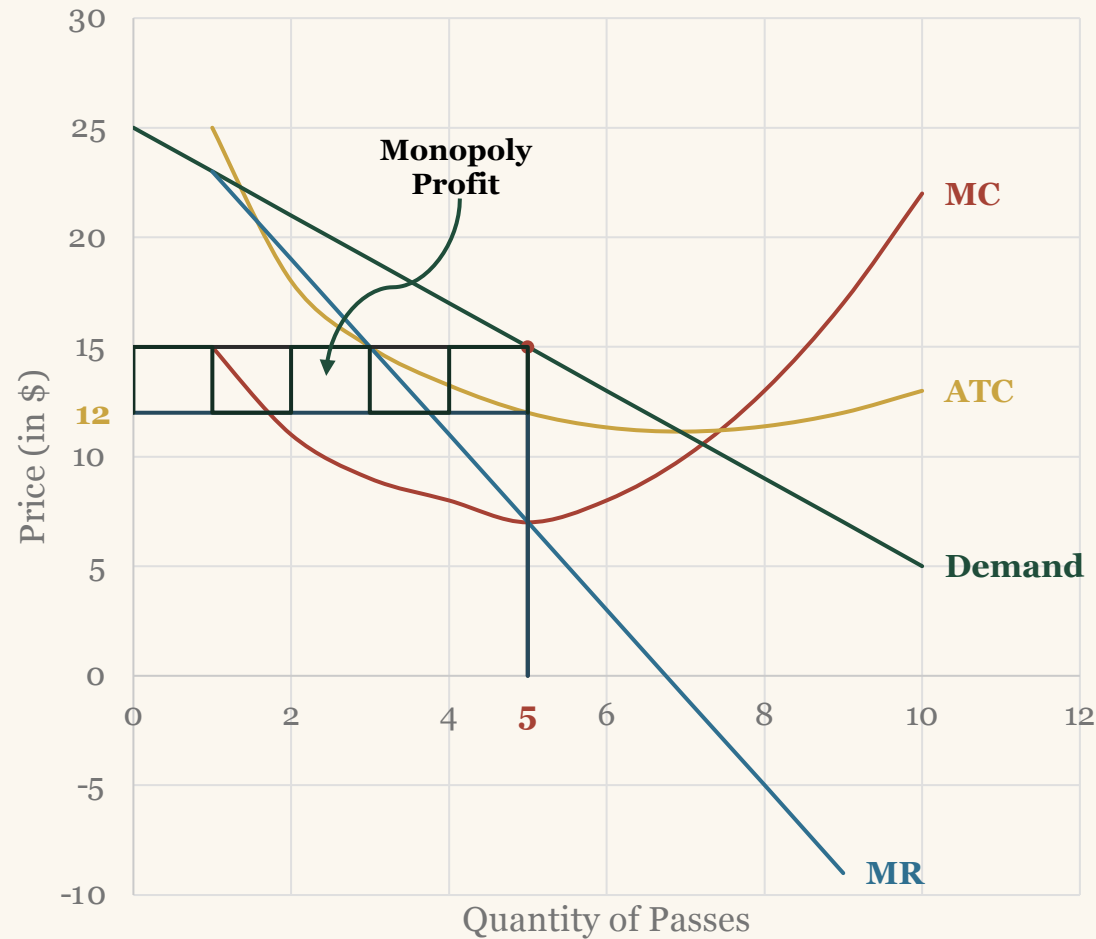


River-Gate chooses output where:
 $MR = MC$

And ATC gives the corresponding
Profit.

$$Profit = (P - ATC) \times Q$$

Monopoly Profit

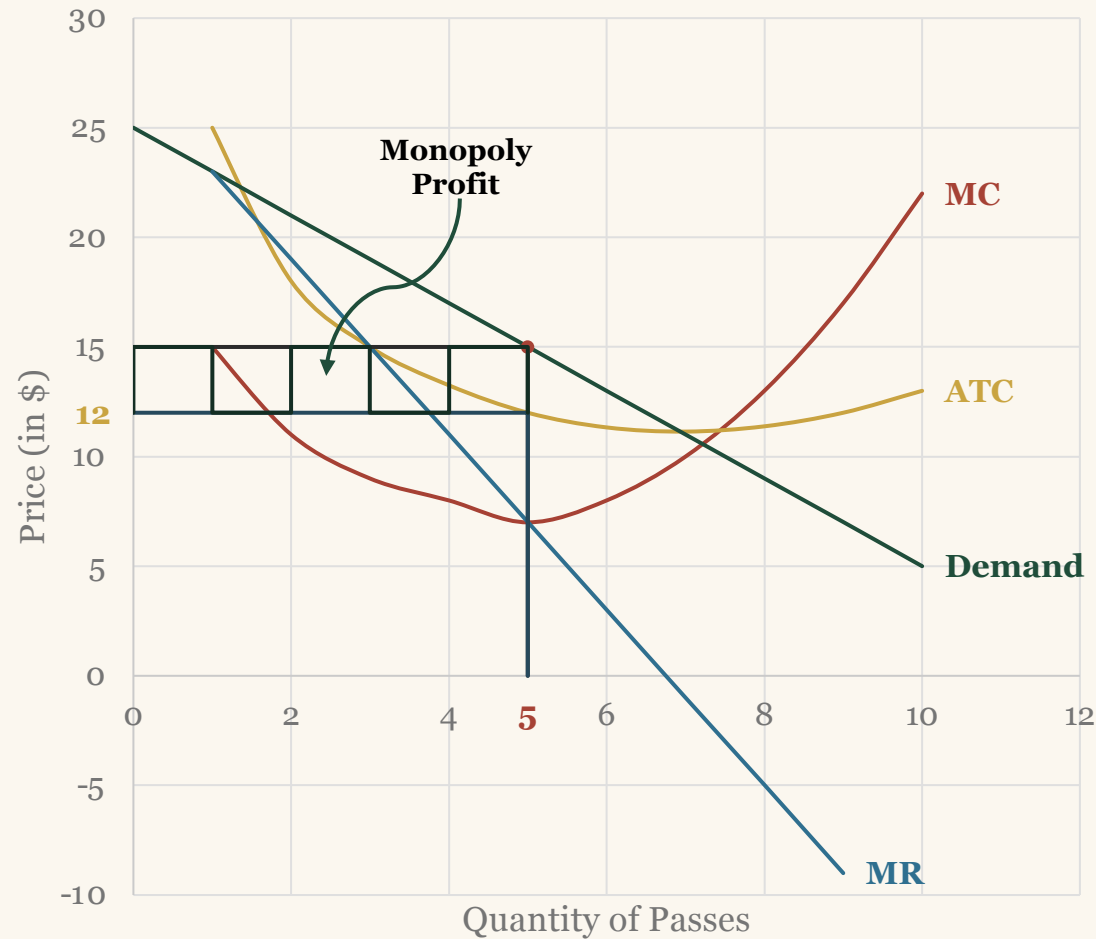


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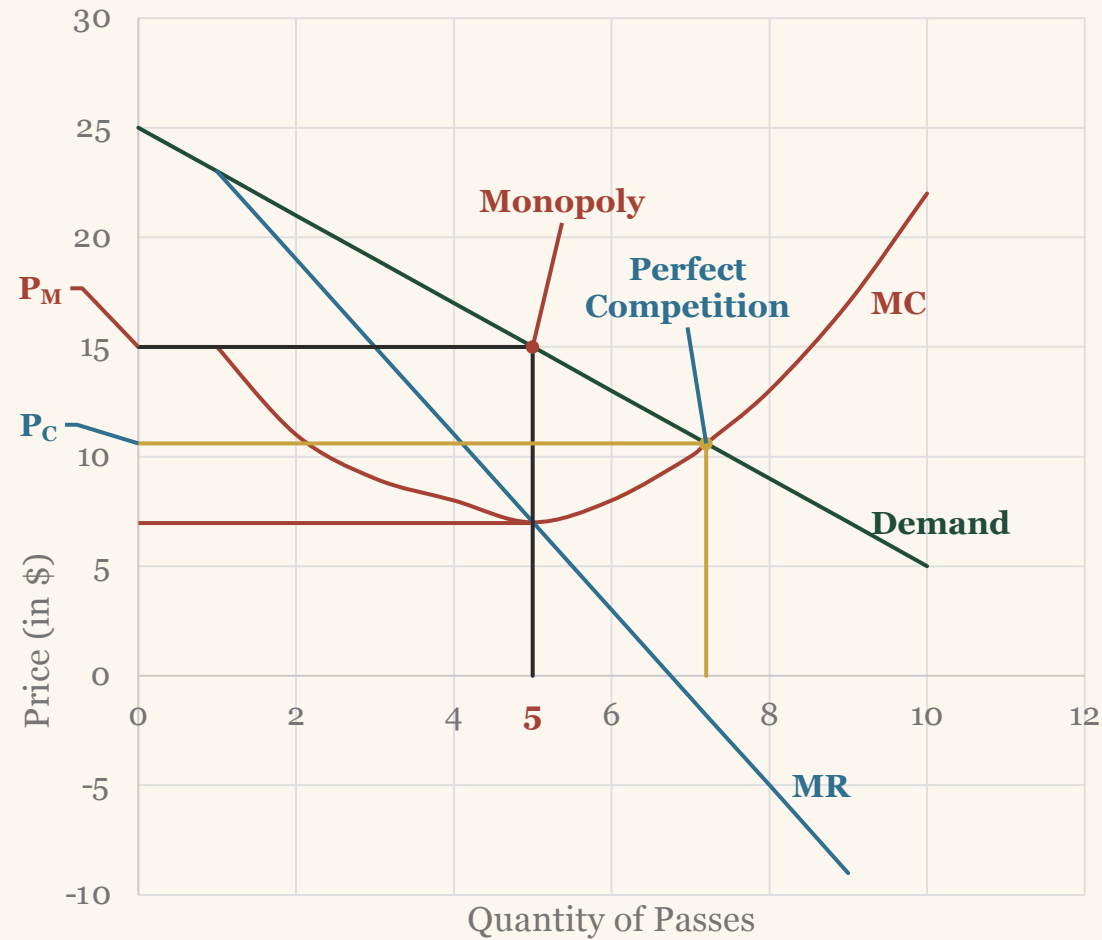
Long Run Profit



There are barriers to entry. No new firms can enter the market.

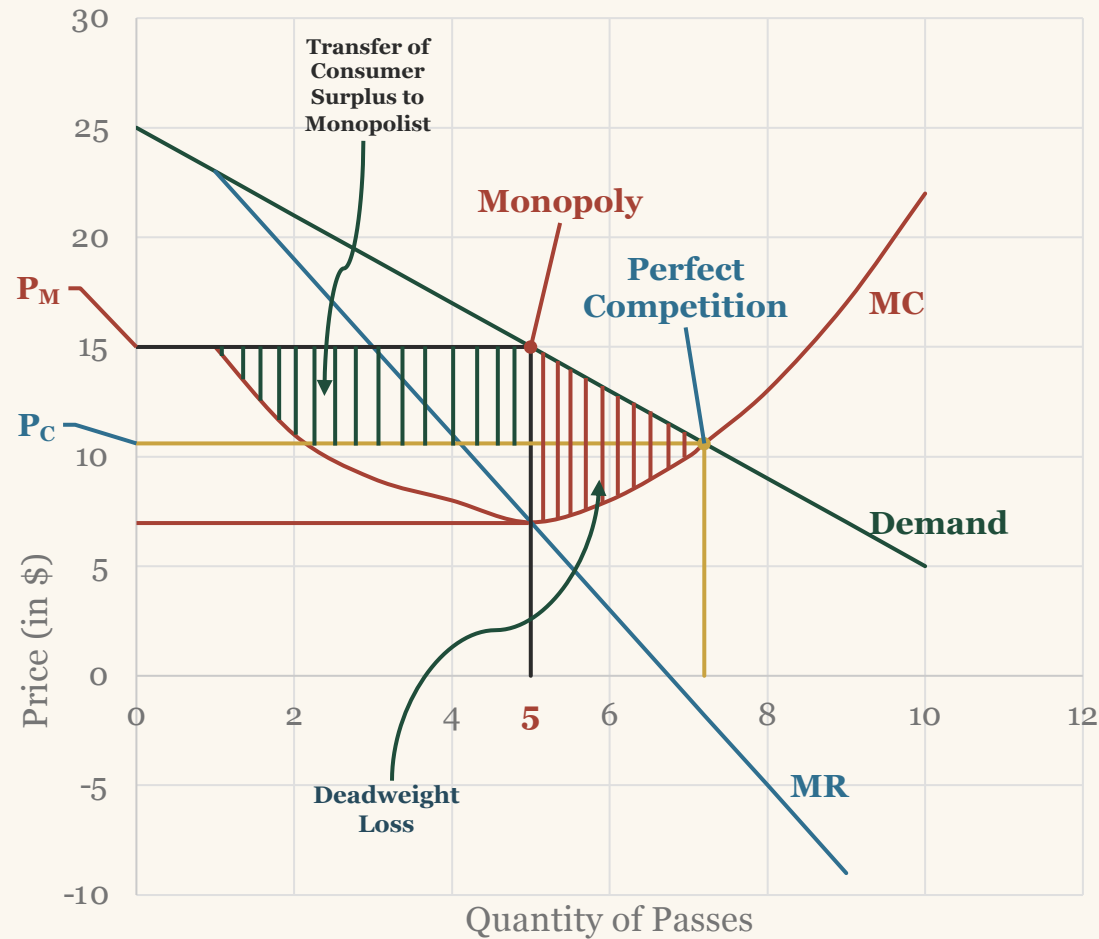
The Long Run profit remains the same.

Inefficiency of Monopoly



At Perfect Competition:
 $P = MC$

Inefficiency of Monopoly



At Perfect Competition:

$$P = MC$$

Monopoly:

$$MR = MC$$

Deadweight Loss is generated

Can River-Gate Charge Different Prices?

So far, River-Gate charged one price to everyone.

But some visitors will pay a lot to reach Lantern Island.

Some residents will go only if the pass is cheaper.

If River-Gate can separate buyers, it may charge different prices.



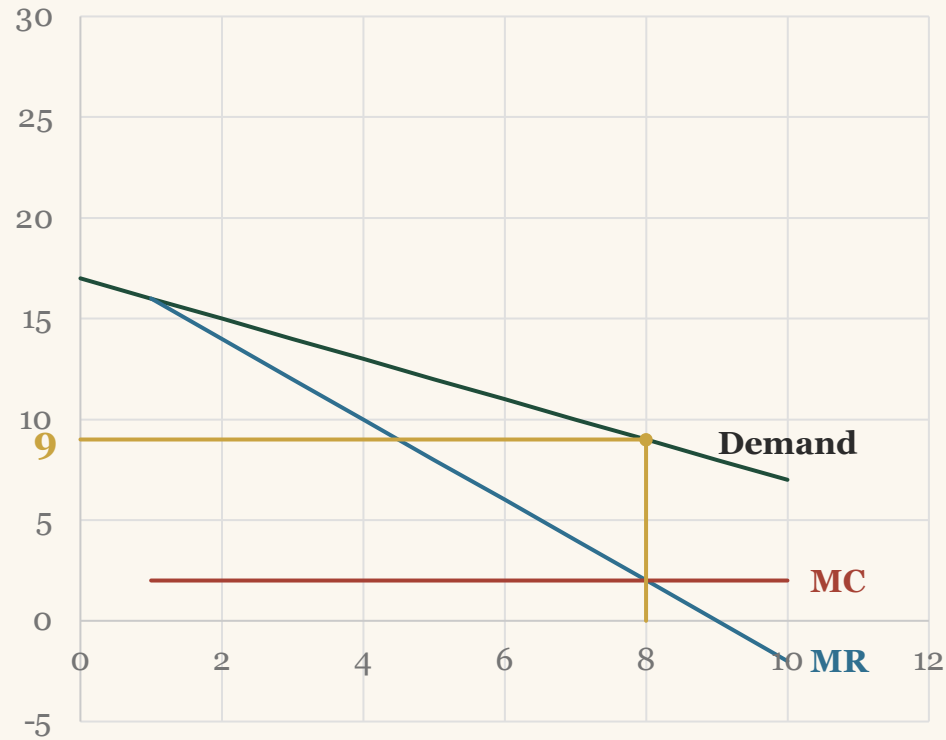
Can they do this?

Price Discrimination

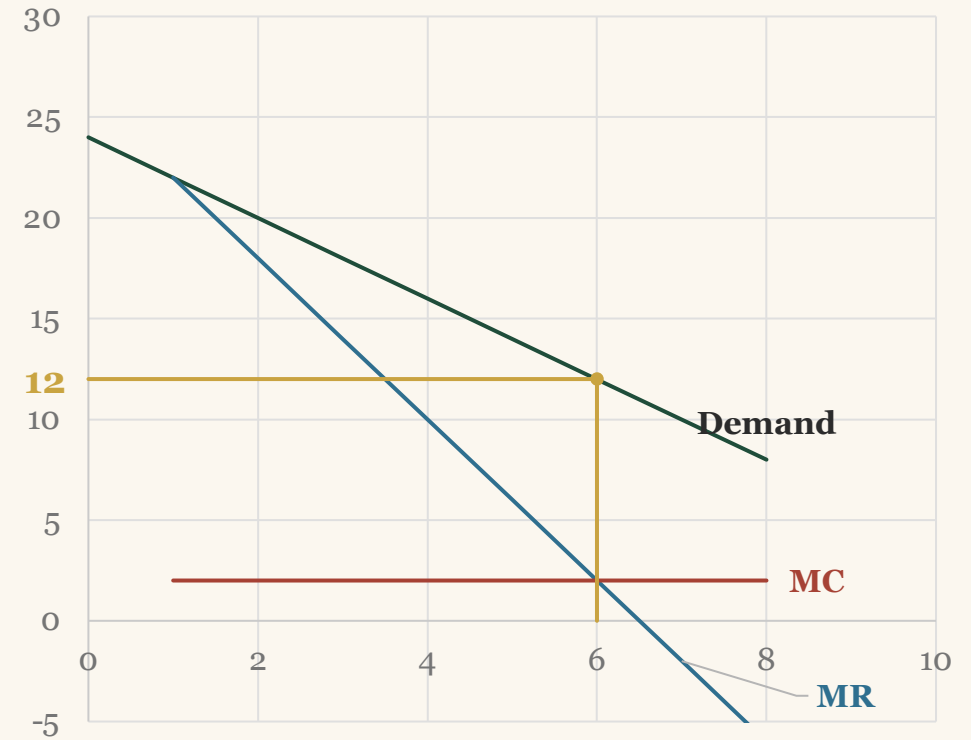
Charging different prices to different customers for the same good or service when the price differences are not due to differences in cost.

Different Prices for Students, Tourists and Residents.

Students vs Tourists

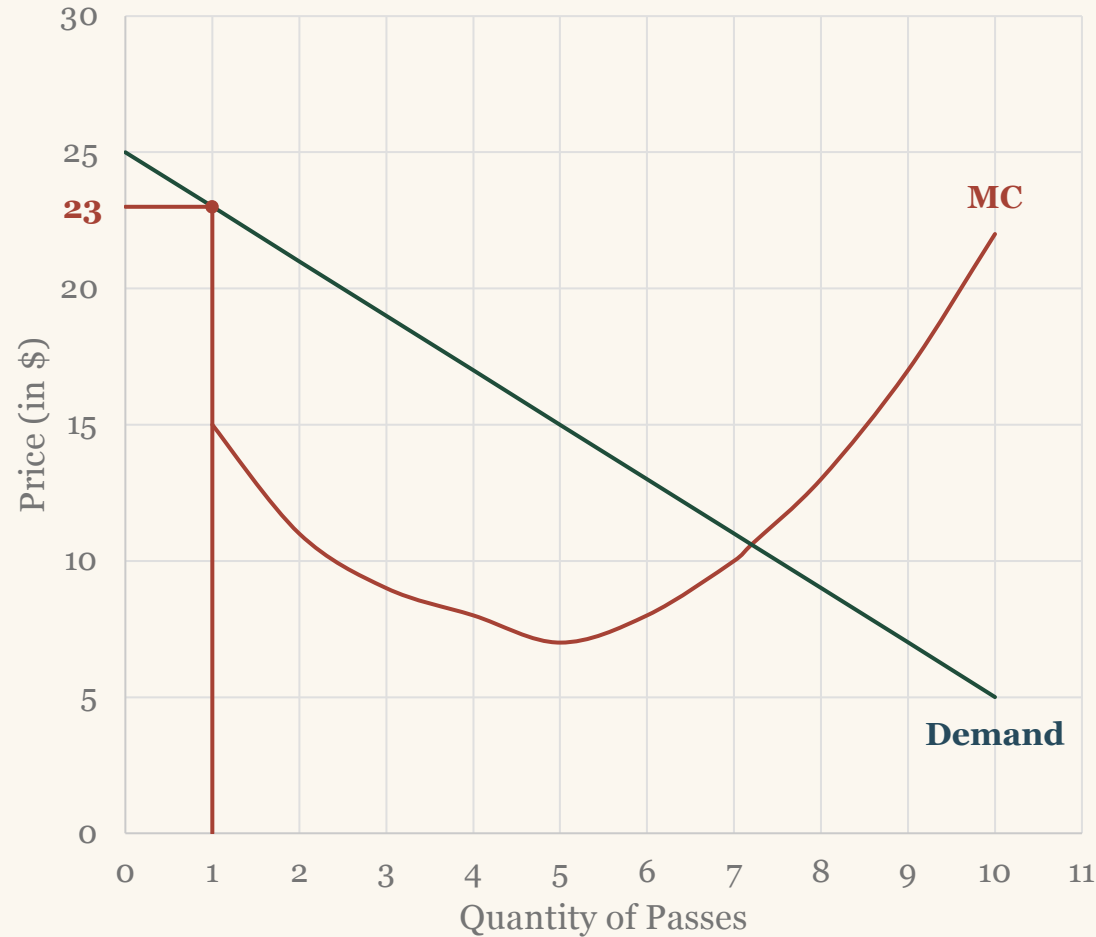


Students
Price = \$9



Tourists
Price = \$12

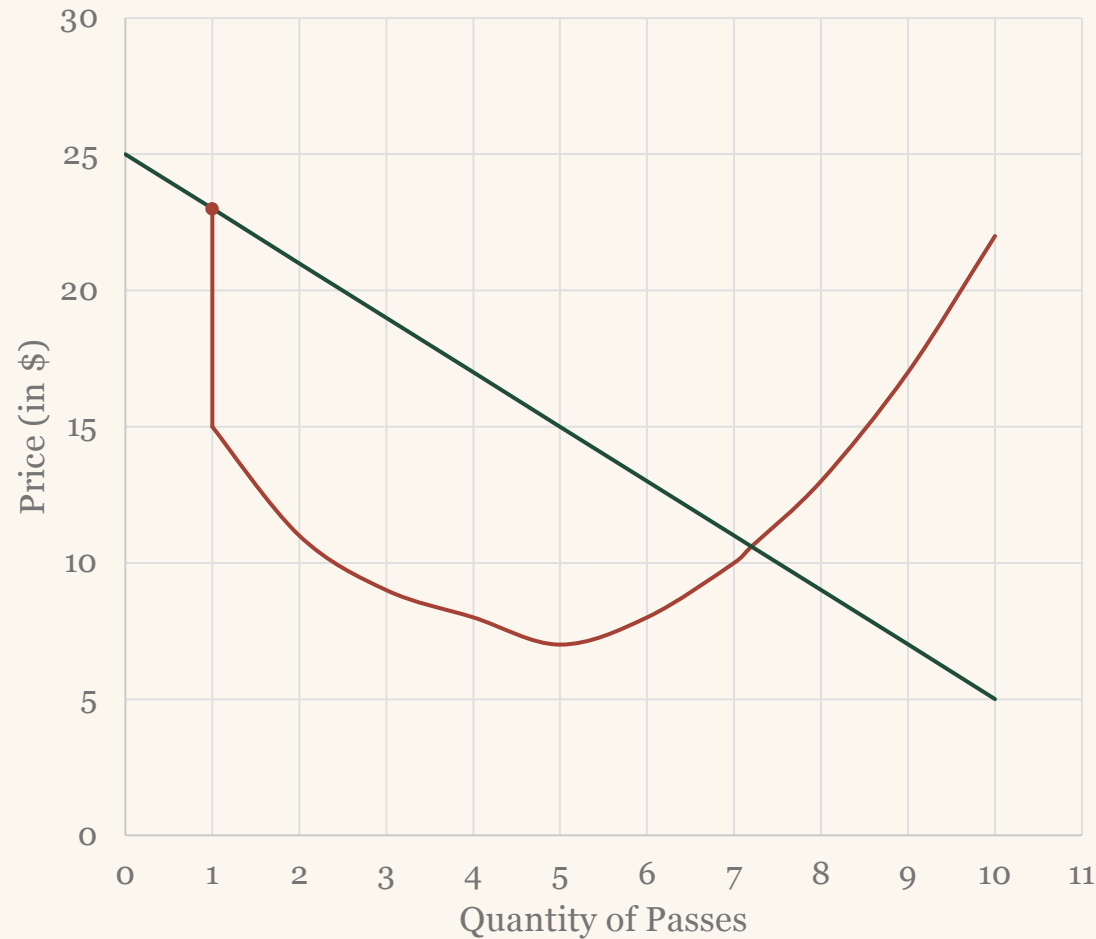
Perfect Price Discrimination



River-Gate knows exactly what each buyer is willing to pay.

Each pass can be sold at a different price.

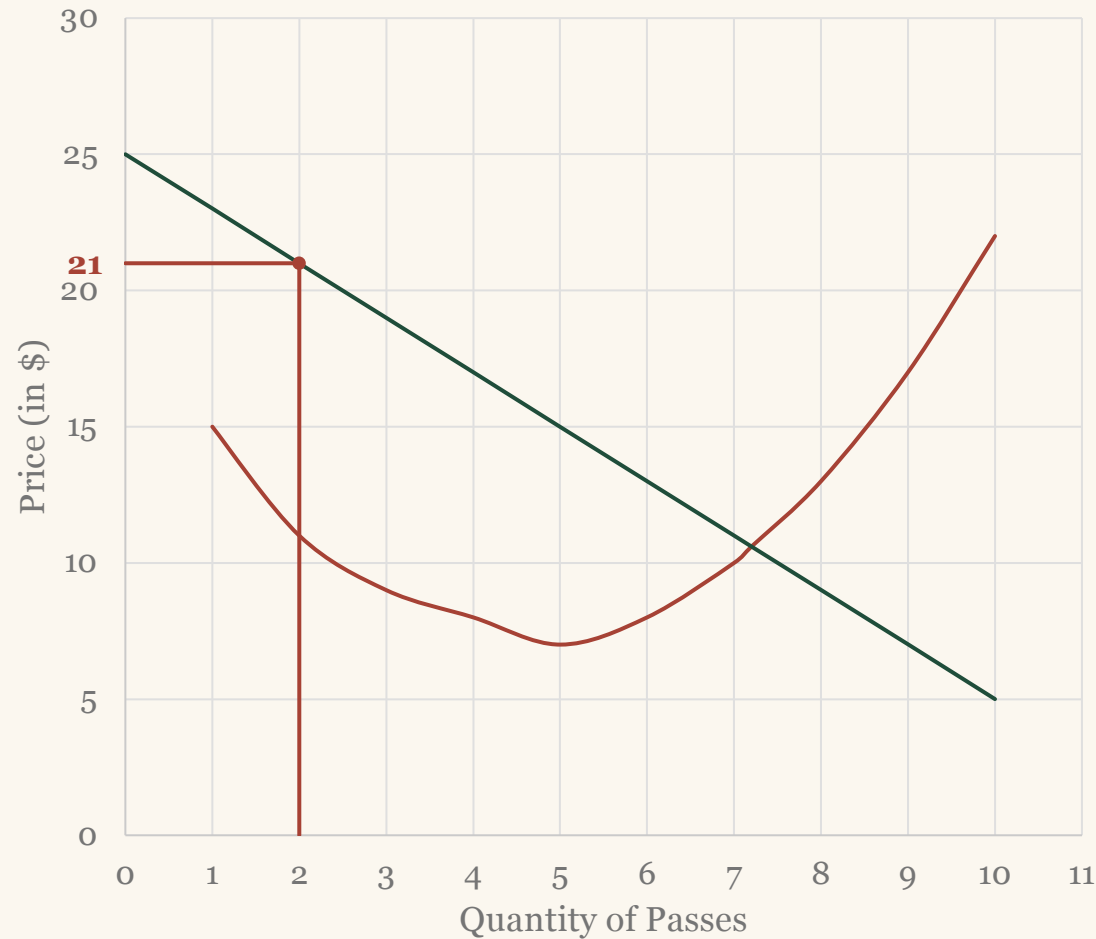
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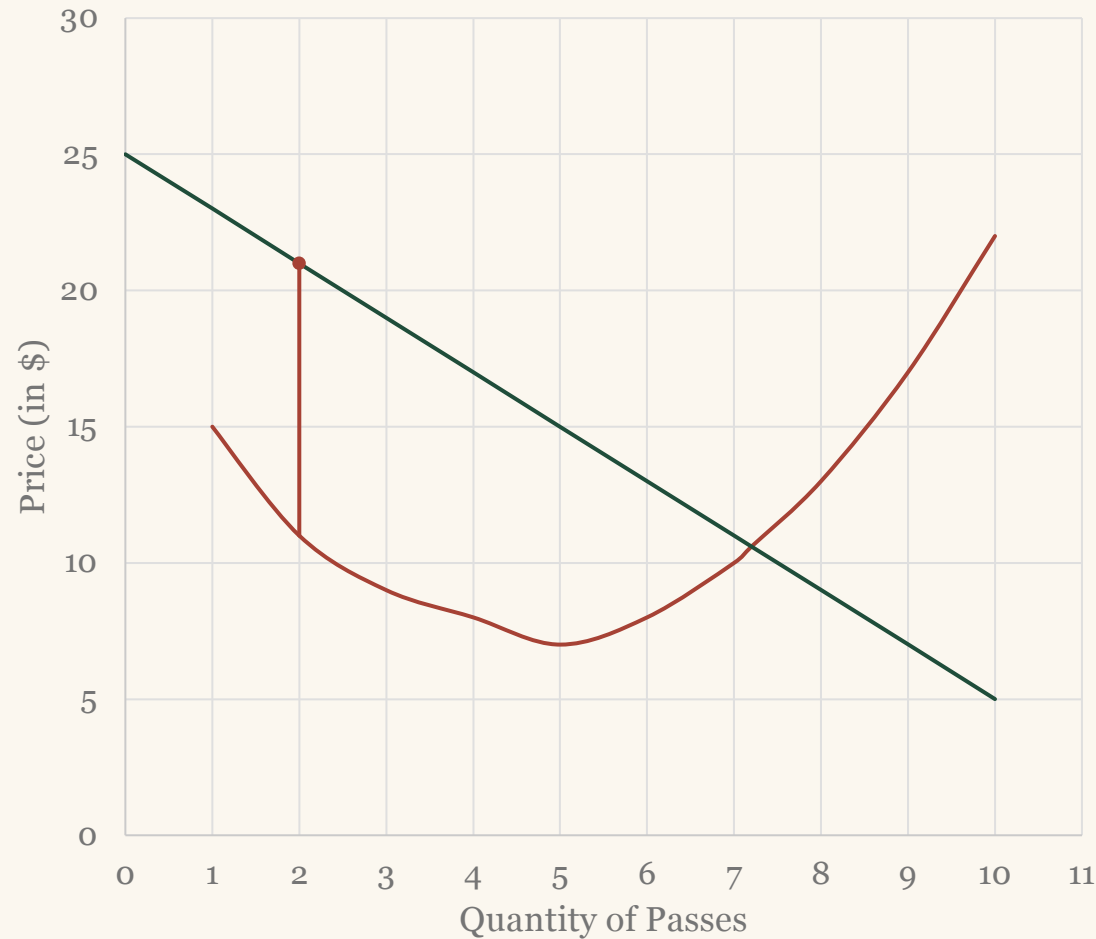
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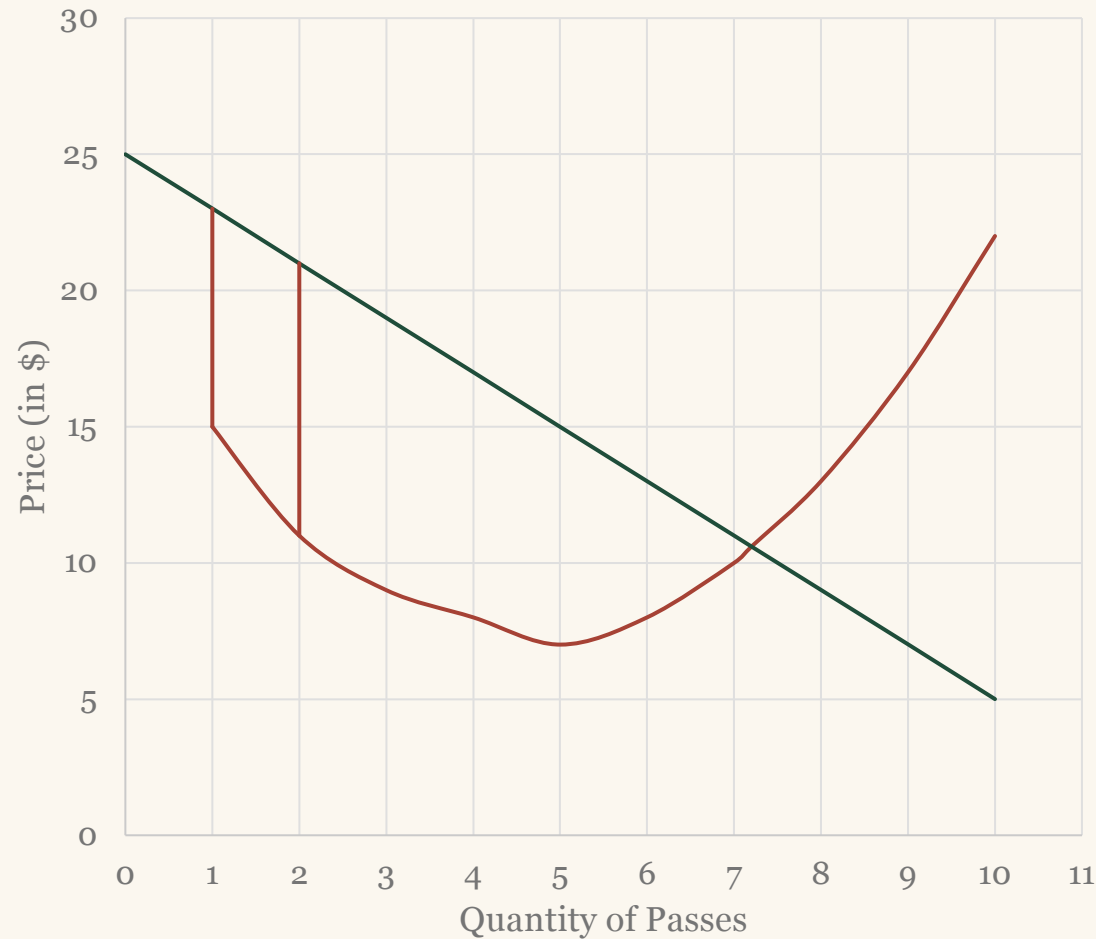
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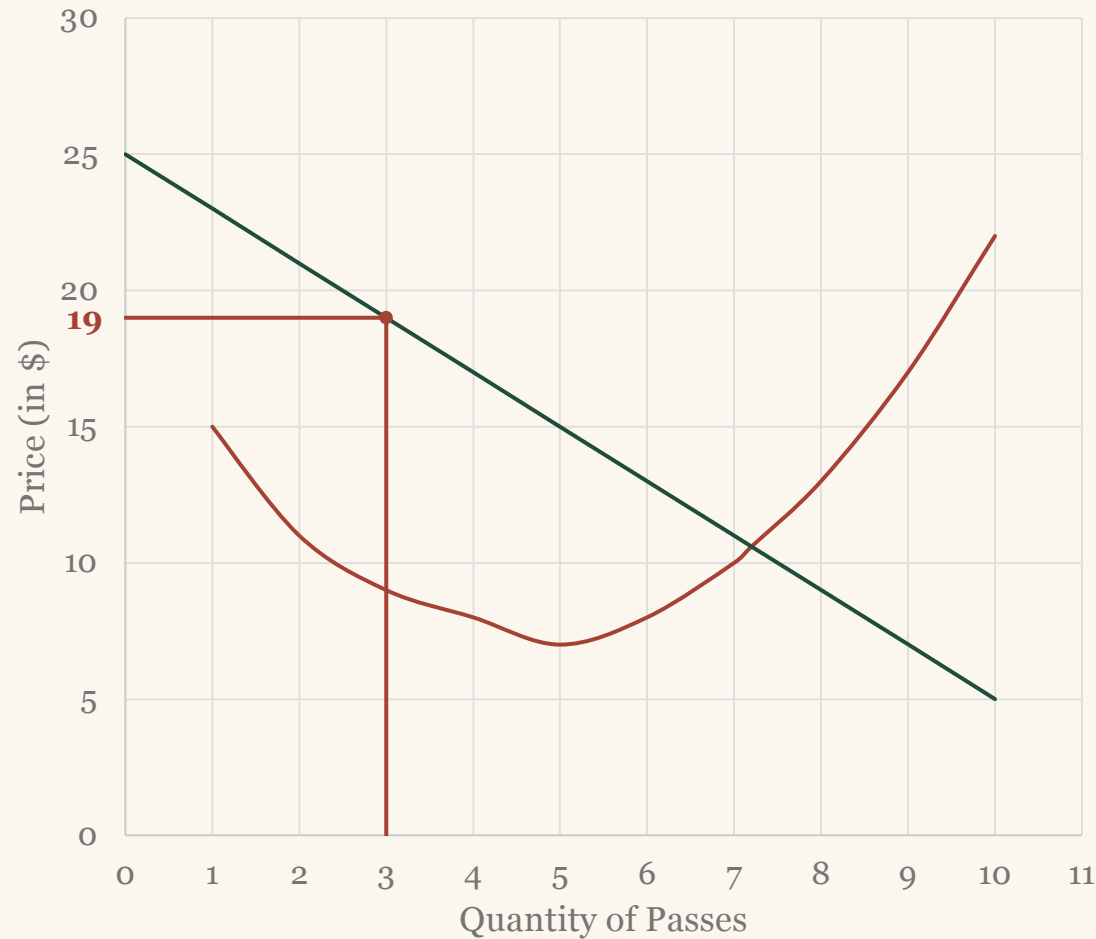
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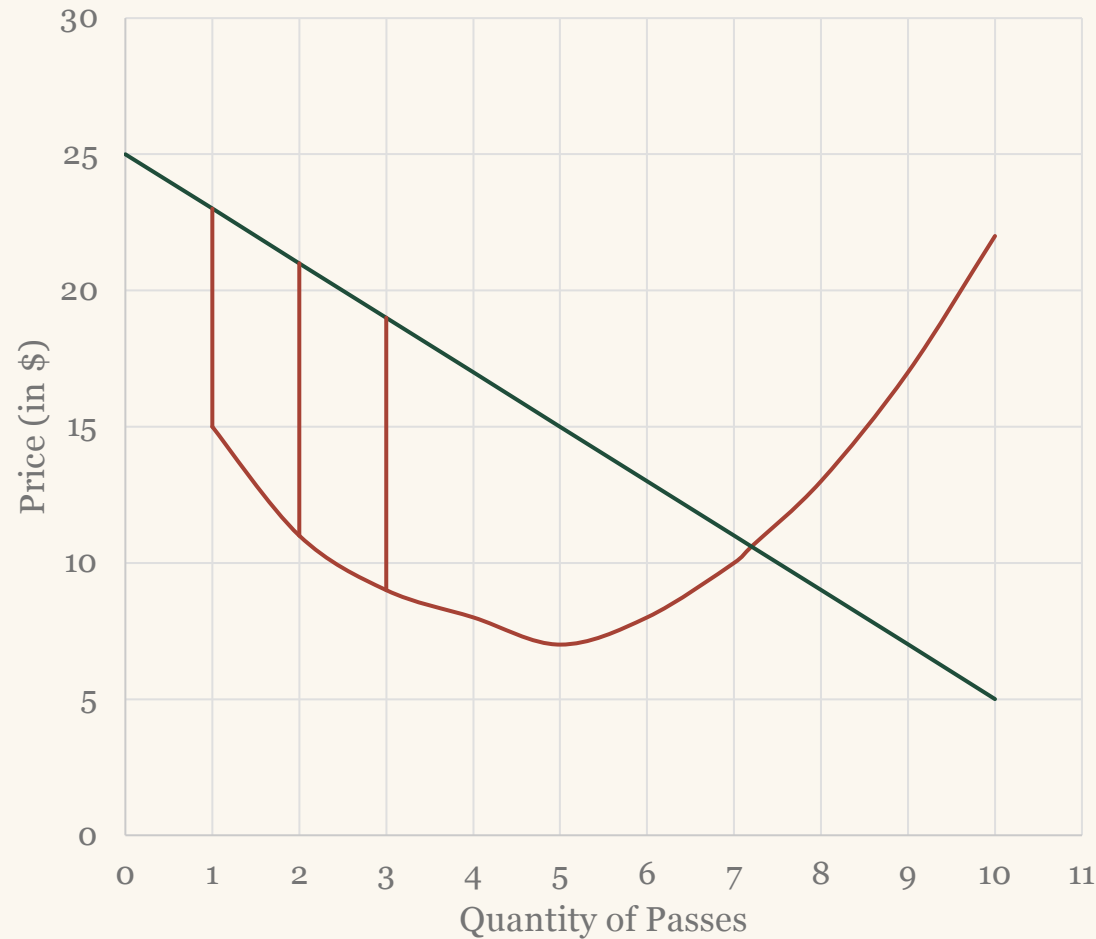
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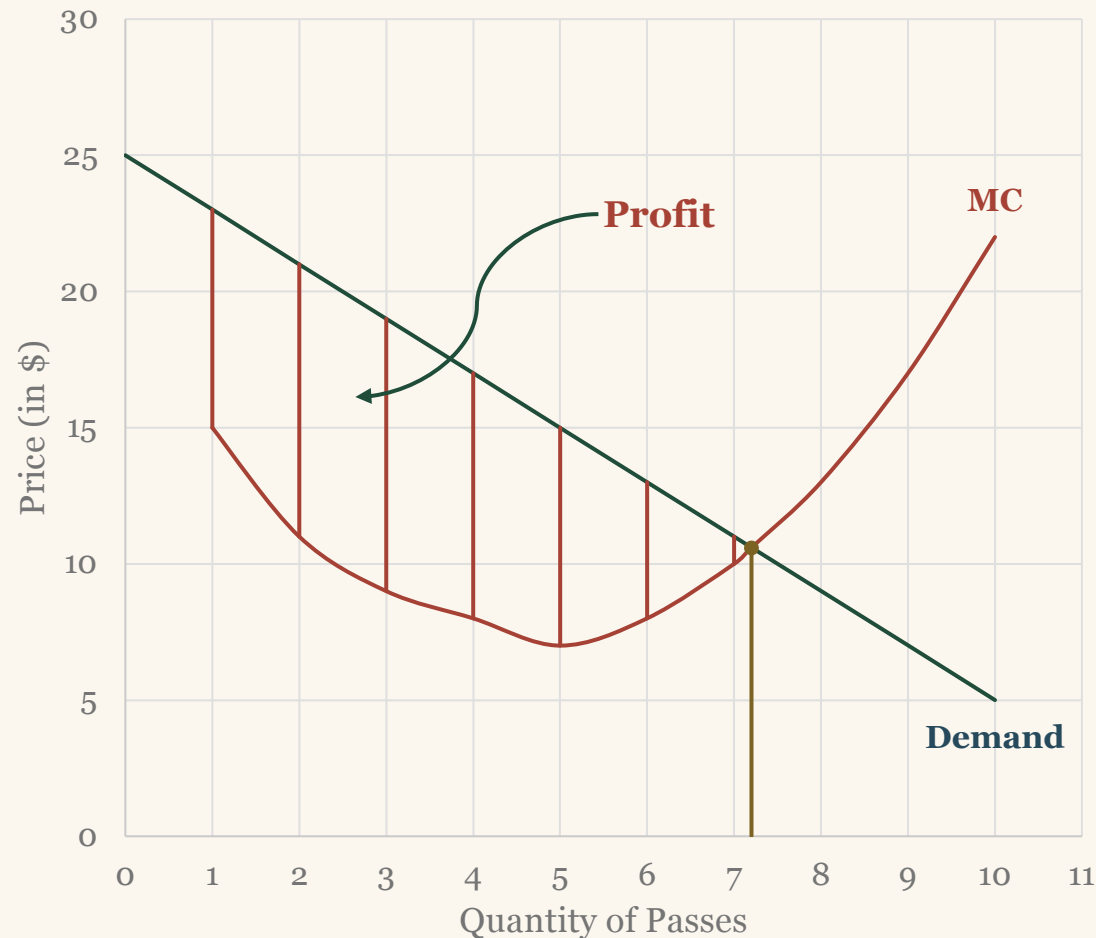
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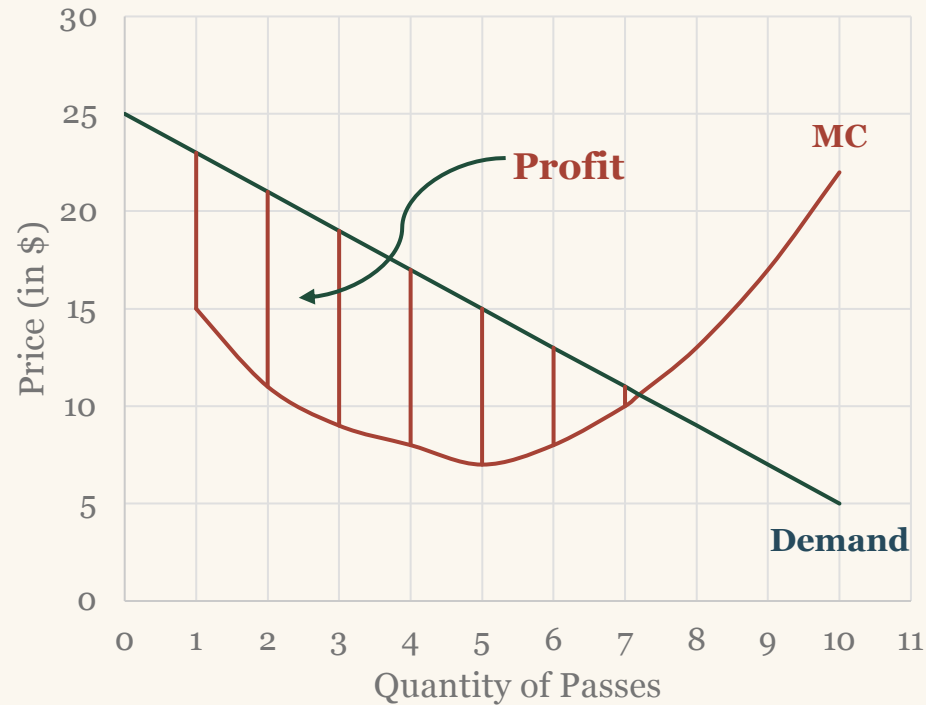
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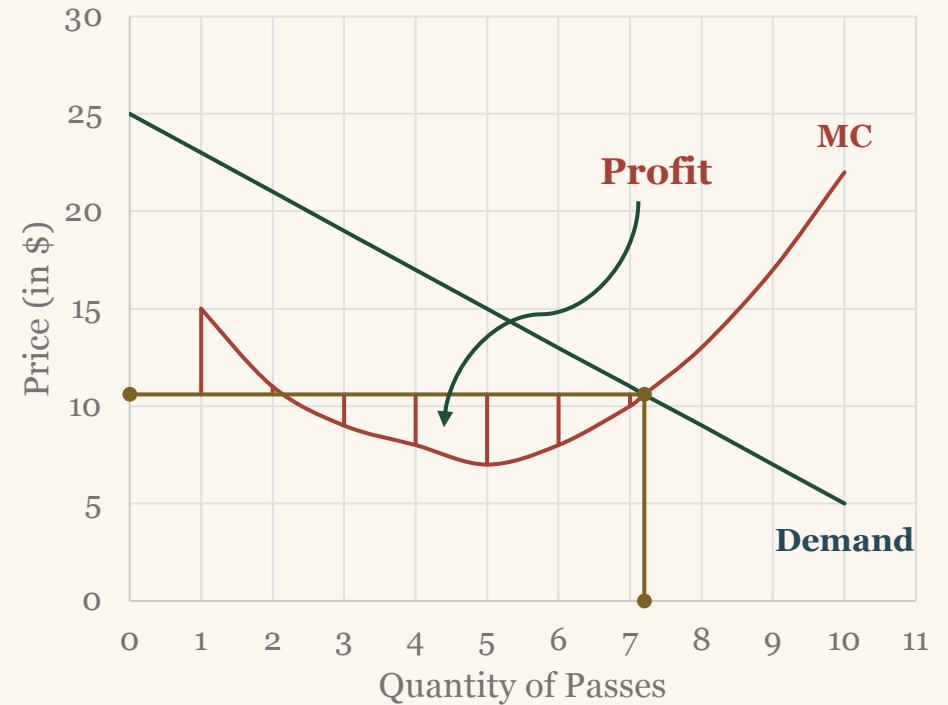
So, it sells until:
 $P = MC$

That is the competitive quantity, but the surplus goes to River-Gate.

Comparing to Perfect Competition



Price Discrimination



Perfect Competition